

K2 - 18b is a hycean (hydrogen rich atmosphere) planet. The planet is covered with hydrogen gas meaning could bring the temperature of the planet at a level where it is liquid but:

Hypothesis 1 (Hycean World): It has a liquid water ocean under a thin Hydrogen/Helium atmosphere with Carbon Dioxide (CO_2) and Methane (CH_4).

Hypothesis 2 (Mini-Neptune): It is a gas planet with no surface, a thick hydrogen envelope, and a deep magma ocean underneath.

We can find the thickness of the hydrogen envelope by calculating the Atmospheric scale height.

Transmission spectroscopy method: When a planet comes in front of its star, the light coming from the star gets a little bit dim (about 2900 parts per million) and the planet's air blocks slightly more or less starlight depending on the color of the light, because different gases soak up different colors.

This transit method happens in both low-res and high-res spectroscopy. We will use the low-res data because the high-res are usually wrong by 200 to 400 PPM.

Q. How heavy the gas molecules in the planet's air are?

Wiggle: small extra dimming at colors where gases absorb, compared with nearby colors where they don't. $\Delta D = (\text{Band Depth} - \text{Baseline Depth})$

Where Baseline depth is the normal amount of light the planet blocks
and Band Depth is amount of light planet blocks when different gases absorbed

The size of the wiggle tells you how tall one "scale height" of air is. A scale height is the height over which the air thins out by a fixed amount, and a light gas puffs up high while a heavy gas stays squashed low. From that height and the temperature and gravity, you can work backwards to the average weight of the gas molecules, which is called μ .

Formula A: Calculate Scale Height (H) from the Spectrum

The atmospheric modulation amplitude (ΔD) relates to scale height by the geometric formula:
(Where $N \approx 4 \text{ to } 5$ scale heights for typical absorption features).

$$\Delta D \approx \frac{2 \cdot R_p \cdot (N \cdot H)}{R_*^2}$$

Formula B: Solve for Mean Molecular Weight (μ)

Now take that calculated H and plug it into the atmospheric physics formula:

$$H = \frac{k_B \cdot T}{\mu \cdot m_H \cdot g}$$

- $\mu \approx 2.0$ — 2.3: Points to a Hydrogen/Helium-rich envelope (Mini-Neptune model).
- $\mu > 3.5$ — 4.0: Points to heavy atmospheric gas enrichment (Hycean ocean-candidate model).

There are two ways to measure the wiggle:

Method A: I took the highest point and subtract the lowest point. When data are noisy, the highest point is partly a lucky upward blip and the lowest is partly a lucky downward blip, so you get a big wiggle even from a flat, boring line. That means Method A was mostly measuring noise. It gave μ values between 1.35 and 1.92, which are too light to be physically possible for hydrogen and helium air, so I flagged them as impossible.

Method B: I picked the places where methane and carbon dioxide are known to absorb light, around 1.15, 1.4, 1.7, 2.3, 3.3 and 4.3 micrometers, and for each one I compared the average depth inside the absorbing zone with the average depth in nearby quiet zones on either side. While carrying uncertainties through each, the six features gave 34 to 77 ppm. They agreed with each other within their errors, and I combined them into one best estimate of about 56 ± 8 ppm. That becomes a scale height of about 40 km and a μ of about 4.3 (± 0.6) if you assume four scale heights, or 5.3 if you assume five. Next I redid it at 140 K instead of 255 K, since newer papers say the upper air is colder than the planet's average temperature, and because μ scales directly with temperature, the answer dropped by 45% to about 2.3 to 2.9.

$$\Delta D = 56.1 \pm 8.4 \text{ ppm}$$

Plugging in 56.1 ppm (which is 5.61×10^{-5}), $R_s = 308,900$ km and $R_p = 16,620$ km gave

$H = 40.3$ km (for $N = 4$) $H = 32.2$ km (for $N = 5$), with error bars of 6.0 and 4.8 km.

Hot gas puffs up and heavy gas sinks, so if I know how tall the air is, how hot it is (255 K) and how hard the planet pulls it down (12.3 m/s^2), I can work out how heavy the typical gas particle must be and the answer was **$\mu = 4.25$** for $N = 4$ and **$\mu = 5.31$** for $N = 5$.

The fractional error in μ equals the fractional error in ΔD , so 15% of 4.25 gives ± 0.63 , and 15% of 5.31 gives ± 0.79 . But I noticed that μ goes as 1 divided by ΔD , which means that if the wiggle is a bit bigger the weight drops and if it is a bit smaller the weight rises, so the true error bars are lopsided. I therefore also computed the "exact" range by putting ΔD plus and minus one error bar into the formula, and that gave 3.70 to 5.00 for $N = 4$ and 4.63 to 6.25 for $N = 5$.

At 255 K and $N = 4$, the answer of 4.25 sits just above 4.0, but its error bar reaches down to 3.70, so it straddles the boundary and dips into the gap, while at $N = 5$ it is clearly in the heavy zone, and at 140 K it slides down to the light boundary (2.33) or into the gap (2.92). So I wrote that the data are consistent with both scenarios, with only a slight lean toward the heavier one, and only if you accept the 255 K assumption.